

LME Metals Risk Premia — Strategy Reference

LME Copper (2006–2025) | LME Aluminium (2006–2026) | June 2026

Common conventions: PnL = position × $\Delta F1_{\text{continuous}}$ (ratio back-adjusted). TC = 5 bps round-trip charged on $F1_{\text{raw}}$ at each flip. Same-Day: position_{t} = signal_{t-1} (shift-1, no look-ahead). Lag-1: position_{t} = signal_{t-2} (signal observed, then executed next day). Sharpe = annualised, active-day convention.

Family / Variant	Signal Formula	Position Rule	Timing	OOS Category
TAB 7 — MOMENTUM				
MA Crossover (grid search)	signal(t) = sign[MA(F1_raw, m) – MA(F1_raw, n)] Best: Cu MA(35,43), Al MA(60,115)	+1 long / –1 short	Lag-1 ★ (Same-Day also tested)	Fixed-param WF
Anchors EW MA(10,25) + MA(35,43) + MA(63,100)	pos_EW(t) = mean[sign(MA_fast,i – MA_slow,i)] for i=1,2,3	{–1, –2/3, –1/3, 0, +1/3, +2/3, +1}	Same-Day (structural)	Parameter-free
Anchors + IS-Opt Weights (same 3 anchors)	w* = argmax Sharpe(w; IS data), w≥0, Σw=1 port(t) = Σ w*_i · sign(MA_fast,i – MA_slow,i)	Continuous (–1 to +1)	Same-Day	Rolling WF (true OOS)
CTA Single-scale (S, L) pairs — grid	z(t) = [F1(t) – F1(t–L)] / σ(F1, S) pos(t) = sign(z(t))	+1 / –1 (no flat)	Lag-1	Fixed-param WF
CTA Baz-Granger Paper 3 speed tiers (published S,L)	z_i = [F1(t)–F1(t–L_i)] / σ(F1,S_i) u_i = z_i · exp(–z_i²/4) / 0.89 pos(t) = mean(u_fast, u_med, u_slow)	Continuous; rescaled to unit variance	Lag-1	Parameter-free
TAB 8 — CARRY				
V1 Roll Yield — level (F1–F2)/F1, (F1–F3)/F1, (Cash–3M)/Cash	signal(t) = (Fa – Fb) / Fa	pos(t) = sign(signal(t)) +1 backwardation / –1 contango	Same-Day ★ (Lag-1 tested)	Parameter-free
V2 Long Slope — 10 tenor pairs F3–F15, F4–F16, ... , F12–F24	signal(t) = (Fj(t) – Fk(t)) / Fk(t)	pos(t) = sign(signal(t)) +1 / –1	Same-Day (Lag-1 tested)	Fixed-param WF
V3 Z-Score of Roll Yield (252d) [Al primary config]	ry(t) = (F1–F2)/F1 signal(t) = [ry(t) – μ_252(ry)] / σ_252(ry)	pos(t) = sign(signal(t)) +1 / –1	Same-Day ★	Fixed-param WF
V4 Carry-Momentum — Δroll yield N = 5, 10, 20, 60 d [Cu: 20d best OOS]	ry(t) = (F1–F2)/F1 signal(t) = ry(t) – ry(t–N)	pos(t) = sign(signal(t)) +1 / –1	Same-Day ★	Fixed-param WF
TAB 9 — VALUE				
V1 MA Reversion (grid search) Fk: F1–F15; LB: 1/3/5/7/10yr; T: ±5–20% Cu best: F8 5yr Al best: F12 5yr	dev(t) = [Fk(t) – MA_N(Fk, t)] / MA_N(Fk, t)	+1 if dev < –T (cheap) –1 if dev > +T (expensive) 0 otherwise (flat ~45% of days)	Lag-1 ★ (Same-Day gives negative Sharpe)	Fixed-param WF
V2 Baz-Granger Reversal (grid search) F1_raw; LB: 1/3/5/7/10yr Best: 10yr (Sharpe +0.51 IS, –0.36 excl. COVID)	rev(t) = F1_raw(t–N) – F1_raw(t) (positive = price has fallen = cheap)	pos(t) = sign(rev(t)) +1 / –1 (no flat zone)	Lag-1 ★	Fixed-param WF
TAB 10 — PORTFOLIO				
Equal-Weight (EW) — Copper Mom: MA(35,43) Carry: CM-20d Value: V1-F8-5yr	port(t) = ⅓·mom_pos(t) + ⅓·carry_pos(t) + ⅓·val_pos(t)	{–1, –2/3, –1/3, 0, +1/3, +2/3, +1} Corr: Mom-Carry +0.07, Mom-Val –0.21, Carry-Val –0.05	Same-Day (all legs)	Fixed-param WF
Equal-Weight (EW) — Aluminium Mom: MA(60,115) Carry: Z-252d Value: V1-F12-5yr	port(t) = ⅓·mom_pos(t) + ⅓·carry_pos(t) + ⅓·val_pos(t)	{–1, –2/3, –1/3, 0, +1/3, +2/3, +1} Corr: Mom-Carry +0.23, Mom-Val –0.36, Carry-Val +0.01	Same-Day (all legs)	Fixed-param WF
Inverse-Vol (63d window) (alternative weighting)	σ_i(t) = rolling 63d ann. vol of leg i's daily returns w_i(t) = (1/σ_i) / Σ_j(1/σ_j), Σw=1	Continuous (–1 to +1) Rebalanced daily	Same-Day (all legs)	Fixed-param WF

Performance Summary — Primary Deployed Configurations

IS = full-history backtest. WF-OOS Sharpe = avg-of-windows (1yr OOS / 5yr IS rolling, starting ~2011, net 5bps TC). ★ Fixed-param: params selected on full history; WF is sub-period decomposition, not a held-out test.

Strategy / Config	Metal	IS Sharpe (gross)	IS Sharpe (net 5bps)	IS Ann Ret (gross)	Max DD	WF-OOS Sharpe (net)	OOS Methodology
Momentum MA(35,43)	Copper	+0.72	+0.70	+18.3%	−64%	+0.46	Fixed-param WF
Momentum MA(60,115)	Aluminium	+0.48	+0.48	+10.7%	−50%	+0.33	Fixed-param WF
Carry Carry-Mom 20d	Copper	+0.52	+0.47	+13.5%	−47%	+0.28	Fixed-param WF
Carry Z-score 252d	Aluminium	+0.66	+0.64	+14.1%	−50%	+0.83	Fixed-param WF
Value V1 F8 5yr ±10%	Copper	+0.28	+0.27	+3.8%	−67%	+0.36	Fixed-param WF
Value V1 F12 5yr ±10%	Aluminium	+0.27	+0.26	+3.1%	−68%	+0.33	Fixed-param WF
EW Portfolio (3 sleeves)	Copper	+1.04	+1.00	+11.9%	−27%	+0.73	Fixed-param WF
EW Portfolio (3 sleeves)	Aluminium	+0.87	+0.85	+9.3%	−21%	+1.03	Fixed-param WF
Anchors + IS-Opt Weights	Copper	—	—	—	—	In progress	Rolling WF (true OOS)

Data source: LME futures curve (F1–F27), LME Cash & 3M prices. Framework: Bouchoev (2024) commodity risk premia decomposition. WF-OOS avg-of-windows convention matches dashboard reporting across all tabs. Inverse-vol trailing window (63d) selected as Sharpe-optimal; EW outperforms IV on net Sharpe and max drawdown.